

Iron deficiency anaemia management in Maternity guideline (GL783)

Approval

Approval Group	Job Title, Chair of Committee	Date
Maternity & Children's Services Clinical Governance Committee	Chair, Maternity Clinical Governance Committee	1 st July 2022 & 2 nd September 2022

Reviewing Pharmacist (only if the document includes medications)	Date
S Horsburgh	n/a

Change History

Version	Date	Author, job title	Reason
6.0	Jan 2022	B Chohan (Consultant Obstetrician), T Hawkins (Transfusion Practitioner), R Sampson (Consultant Haematologist) R Blakely, AN Services manager	Reviewed and updated to reflect current practice and national advice.
6.1	Aug 2022	T Hawkins, Transfusion Practitioner H Inkster, OASI MW S Philip, Consultant O&G	Live change to amend the dosing table to reflect change to single dose from twice daily (pg 4)

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Overview: *Iron deficiency remains a significant problem for pregnant women in the UK. The objective of these guidelines is to provide healthcare professionals with recommendations for the prevention, diagnosis and treatment of iron deficiency in pregnancy and in the postpartum period.*

Introduction

Early detection and appropriate management of iron deficiency anaemia may prevent otherwise young and healthy patients from receiving an unnecessary blood transfusion.

Patients with Thalassaemia who are diagnosed with iron deficiency anaemia should be reviewed by a haematologist for appropriate management and treatment. Patients with Thalassaemia or sickle cell disease should NEVER routinely receive iron therapy either oral or intravenous.

In the absence of any other diagnosis of anaemia, iron deficiency in pregnancy is defined as:

- Haemoglobin < 110 g/l in 1st trimester
- Haemoglobin < 105 g/l in 2nd and 3rd trimester
- Serum Ferritin < 15µg/l (Most reliable test in pregnancy, consider oral iron supplementation if ferritin <30 µg/l)
- MCH < 25 pg and
- MCV < 80fl
- Transferrin saturation <15%

Recommendation:

An individual approach is preferable, based on results of blood count screening tests as well as identification of women at increased risk. Routine Iron Supplementation is not recommended in pregnancy.

Check Ferritin level for women at booking at risk of IDA

- Known anaemic women where estimation of iron stores are needed (Hb<110g/l)
- Known Haemoglobinopathy (Non-anaemic women with high risk of iron depletion), these patients normally have an anaemia, but can have a microcytosis which makes it difficult to judge if iron deficiency or the haemoglobinopathy is causing the anaemia
- Previous anaemia
- Multiparty P3 and above
- Consecutive pregnancy <1year following delivery
- Vegetarians

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- Teenage pregnancies
- Recent history of bleeding
- Post Bariatric surgery- also check B12 and folate
- History of inflammatory bowel disorders
- Coeliacs

Non-anaemic women where estimation of iron stores is necessary

- High risk of bleeding (Placenta Praevia, Placenta Accreta)
- Jehovah's witness

Recommendation:

- Ensure FBC and Ferritin done at booking appointment in the above group of women and results are checked with in 7-10 days and oral iron therapy is started, if Haemoglobin < 110/l and/or Ferritin < 30 µg.
- Clearly document in clinical notes the baseline ferritin level, HB and start date and dose of oral iron commenced.
- Recheck Haemoglobin level at 16 weeks in women who are on oral iron therapy if there is no improvement in haemoglobin levels consider compliance issue and offer alternative regimen. If there is no compliance issue check red cell folate & Vitamin B12 level and refer patient to secondary care.
- Recheck Haemoglobin level at 28 weeks' gestation and commence oral iron therapy for women with HB < 105 gm/l. If anaemia is diagnosed prior to 28 weeks and anaemia is worsening check compliance and consider alternate regimen to improve compliance.
- Repeat Hb and ferritin at 34 weeks in women diagnosed with Iron deficiency anaemia ONLY. Consider IV iron therapy over oral iron in women with Hb < 80 & no increment in haemoglobin with oral iron therapy of if women > 34 weeks' gestation.

Oral Iron Therapy:

Recent studies showed that fractional absorption of iron in iron-depleted young pregnant woman is maximised by taking elemental iron doses of 40 – 80mg once per day or alternate days. (Pavord et al 2019). This decreases the gastrointestinal side effects and can increase compliance. There is now a leaflet on how to take oral iron available from the Trust website.

All women should be given dietary information both at consultation and in written form to maximise iron intake and absorption Oral iron should not be taken after food or tea/coffee

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as this will decrease absorption. Following adequate treatment, the haemoglobin should rise by 0.1-0.2g/dl per day).

Iron salt	Dose per tablet	Elemental Iron per tablet
Ferrous Fumarate	210 mg daily / alternate days as directed	65
Ferrous Gluconate	300 mg BD daily/ alternate days as directed	35
Ferrous Sulphate	200 mg BD daily/ alternate days as directed	60
Ferrous Feredetate (Sytron)	190mg / 5ml elixir daily/ alternate days as directed	27.5 mg / 5ml

Combined Oral Iron and Folate preparations	Iron salt and dose per tablet	Elemental iron content per tablet
Galfer	Sulphate 300 mg	47 mg
Fefol A	Fumarate 305 mg	100 mg

Treatment with oral iron should continue for 3 months after a normal level has been reached to replenish iron stores. This should be passed on to Community midwives and GPs to continue prescriptions postnatal period.

Intravenous Iron therapy- Indications for use:

Intravenous iron should be used for both antenatal and postnatal women where the convenience weight adjusted single dosing regimen can be advantageous. IV iron is indicated in the treatment of iron deficiency anaemia where oral iron treatment has failed or is unlikely to be effective. For instance, where oral iron is not tolerated due to gastrointestinal side effects, for patients where there is poor compliance or where a reliable rapid increase in haemoglobin is required. Intravenous iron is a more suitable management strategy for patients with moderate anaemia (Hb 80-100g/L) due to iron deficiency than blood transfusion as it is safer, more cost effective and of greater benefit to the patient than transfusion. Please ensure that Hb, ferritin, red cell folate & Vitamin B12 levels have been checked within the last 2 weeks to ensure that Iron supplementation is appropriate. If asymptomatic Hb >70 – <80g/l, IV iron infusion should be considered over blood transfusion.

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Ferinject

Ferinject is a Ferric Carboxymaltose infusion which can be used both in pregnancy and postpartum for treatment of iron deficiency anaemia. Ferinject can be given as a single dose of up to 1000mg over 15 minutes which makes it suitable for out-patient administration.

The dose of Ferinject required may be easily calculated, with a post treatment target of 110g/l: This can be seen on the Power plan on EPR

The cumulative dose required for Hb restoration and repletion of iron stores is calculated by the following Ganzoni formula:

Cumulative iron deficit to be replaced [mg] = 500mg + (weight in kg x (target Hb - actual Hb /10) x 2.4)

Thus a patient who weighs 70 kg, with Hb of 80g/l will need;

$500\text{mg} + (70 \times (110-80/10) \times 2.4)$ grams of Ferinject. = 1004mg

For patients < 66 kg: calculated cumulative dose to be rounded **down** to nearest 100 mg

For patients > 66 kg: calculated cumulative dose to be rounded **up** to the nearest 100 mg.

Thus the example patient should be prescribed 1100mg Ferinject.

Dosage Table

	Increase in Hb required (g/l)				
Body Weight (Kg)	10	20	30	40	50
50	6	7	8	9	11
55	6	7	8	10	11
60	6	7	9	10	12
65	6	8	9	11	12
70	7	9	11	12	14
75	7	9	11	13	14
80	7	9	11	13	15
85	8	10	12	14	16
90	8	10	12	14	16
95	8	10	12	15	17
100	8	10	13	15	17

Dose of Ferinject = number of 100mg vials (see table above) for required Hb increase
NB: Maximum dose Ferinject = 1000mg / week (15mg/Kg) as a single dose

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A repeat FBC should be taken 7-14 days after adequate treatment is given. This should show a rise in Hb in most women, but a rise in MCV and MCH is indicative that treatment is working and iron is being taken up into the red blood cells. A repeat FBC should then show improvement in Hb after a further 7 days. Serum Ferritin should also be taken to indicate the state of the iron stores and whether further treatment is required to replenish these.

Side Effects

The most common side effects are

- Headache (3%)
- gastrointestinal symptoms
- A rash

Anaphylaxis is very rare making it a safe preparation to use, though it is recommended that hydrocortisone, chlorpheniramine and adrenaline be available in case of a severe reaction. These are included on the Powerplan for ease of access

Ferinject may be prescribed by all doctors. It can be drawn up and administered by midwives, and should be given in Day Assessment Unit, on Delivery Suite or Marsh and Iffley Wards.

POSTNATAL IRON THERAPY

Postpartum use of Ferinject may be indicated and may avoid transfusion in some cases. The dose may be calculated in the same way and with treatment; increases in Hb of 20-30g/l by day 14 may be achieved.

Ferinject® Administration

Equipment required:

Give by IV infusion via an infusion pump up to a maximum single dose of 1000mg. Doses of 100mg to 200mg in 50mL of sodium chloride 0.9% (no recommended administration time).

Doses of 200mg to 500mg in 100mL of sodium chloride 0.9%, administer over 6 minutes. Doses 500mg to 1000mg in up to 250mL of sodium chloride 0.9%, administer over 15 minutes

- 1 x steret
- 1 x 21 gauge butterfly (blue)
- Tape or cannula dressing
- Vacutainer blood bottles
- FBC

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- 2 x white needle
- 1 x 5ml syringe
- 5mls Normal Saline (to flush cannula)
- 1 x 10ml syringe
- Ferinject doses up to 1000mg
- Normal Saline 50mls (volume according to Ferinject dose)
- Giving set (appropriate giving set if using a pump)
- Gauze

Procedure for administration of Ferinject® infusion:

Baseline observations:

- Prepare infusion of Ferinject®
- Prepare skin with alcohol steret
- Insert 22 gauge venflon (blue)
- Secure cannula in position with tape or cannula dressing
- Take blood samples from cannula
- Flush with 2mls Normal Saline
- Connect infusion of Ferinject® and infuse at pump rate of 210 ml/hr

Observe patient for any adverse events

Remove cannula, apply pressure, extend and elevate the arm

Post injection observations:

Intravenous Chlorpheniramine, Hydrocortisone and Adrenaline should be available for immediate use in the event of a severe adverse drug reaction.

Patients must be informed that oral iron should be discontinued 24 hours before infusion of intravenous iron and not to take oral iron for at least 5 days after the last infusion of intravenous iron.

- FBC can be rechecked at 6 weeks postnatal visit at GP surgery. Oral Iron therapy should be continued 5 days after infusion until 6 weeks postnatal FBC is done at GP surgery and continued if need be by GP.

Follow Up

Recommendations for treatment with iron or blood usage

- If Hb < 70g/l, transfuse. Aim to increase Hb to 70g/l only (even 1 unit is OK).
- If postnatal Hb ≥ 70 g/l and patient is not at significant risk of further haemorrhage, consider IV iron if symptomatic or oral Iron therapy if asymptomatic.
- If post transfusion Hb ≥ 70 g/l. consider IV iron if symptomatic or oral iron if asymptomatic.

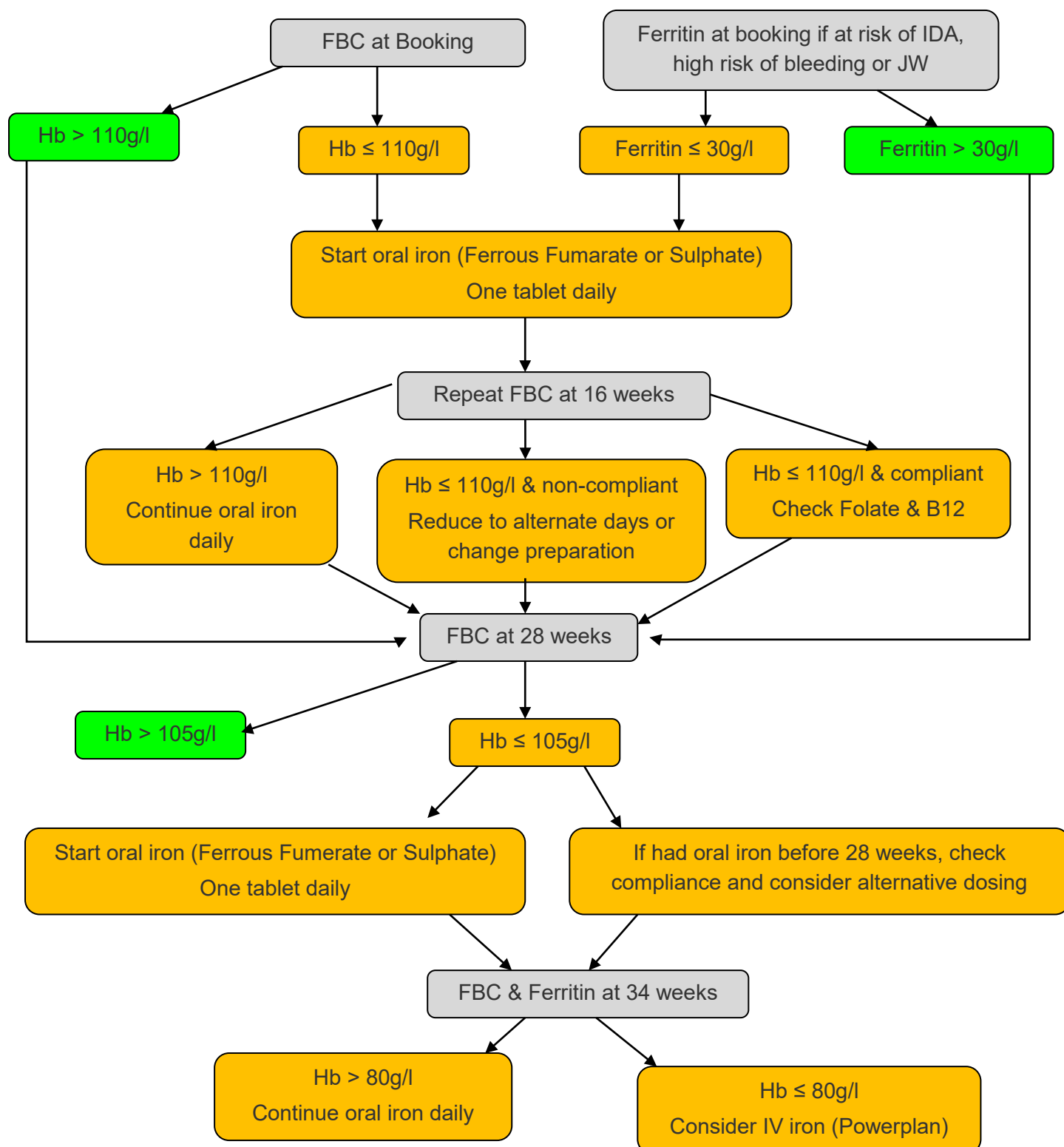
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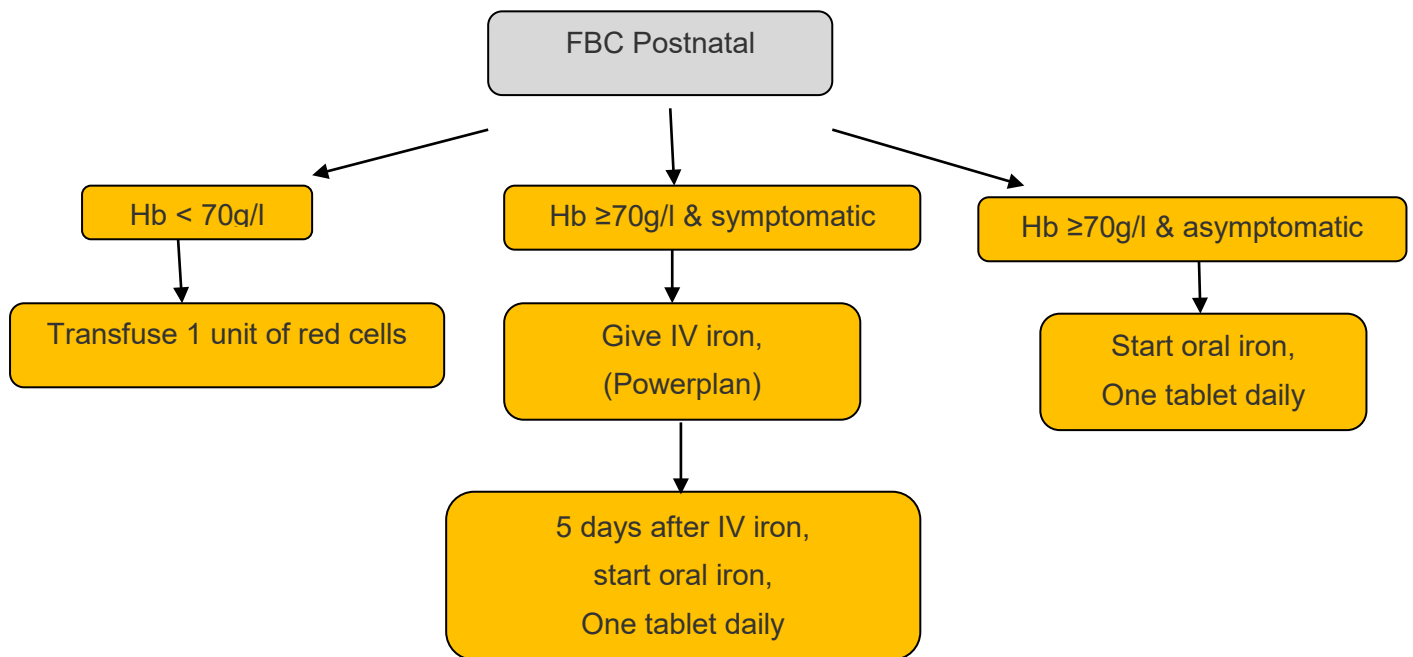
1. Pavord S, Myers B, Robinson S, Allard S, Strong J, Oppenheimer C: UK guidelines on the management of iron deficiency in pregnancy: Br J Haematol 2012;156(5):588-600.
2. Pasricha, S., Flecknoe-Brown, S., Allen, K. (2010) Diagnosis and management of iron deficiency anaemia: a clinical update. Medical Journal of Australia 193, 525-532.
3. National Institute for Health and Clinical Excellence (NICE) guideline Antenatal care on uncomplicated pregnancies. CG62; March 2008.
4. South West Regional Transfusion Committee. Regional template/guideline for the management of anaemia in pregnancy and postnatally. April 2014
5. Bayoumeu, F et al. 2002. Iron therapy in iron deficiency anaemia in pregnancy: intravenous route versus oral route. Am J Obstet&Gynae;186(3).
6. Bhandal, N.2006. Intravenous versus oral iron therapy for postpartum anaemia. British Journal of Obstetrics & Gynaecology, 113(11), 1284-1252.
7. Intravenous iron and serious hypersensitivity reactions: new strengthened recommendations to manage and minimise risk. MHRA Drug Safety Update, Volume 7, Issue 1, August 2013.
8. Reveiz, L., Gyte, G.M.L., Cuervo, L.G. (2007) Treatments for iron-deficiency Anaemia in pregnancy. Cochrane Database of Systemic Reviews 2
9. Royal College of Obstetricians and Gynaecologists (RCOG) Blood Transfusion in Obstetrics. Green-top Guideline No. 47; May 2015
10. Royal College of Obstetricians and Gynaecologists. (2009) Postpartum Haemorrhage, Prevention and Management. RCOG Green-top guideline 52.

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EMA092a - Flowchart for management of antenatal iron deficiency anaemia



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EMA092b - Flowchart for management of postnatal iron deficiency anaemia

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